A Novel Approach to MI-EEG Classification via 3D Interpolation and 3DCNN

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Abstract. Electroencephalography (EEG) contains rich information about brain activity. Classification based on motor imagery EEG (MI-EEG) is essential for active intelligent rehabilitation using brain-computer interface technology. However, most of the current MI-EEG decoding methods are dedicated to the study of efficient time-frequency feature extraction from the raw EEG, ignoring the spatial features associated with electrode distribution. Therefore, this paper proposed an MI-EEG classification method using a combination of three dimensional (3D) spatial interpolation and 3D convolutional neural network (3DCNN) to fully utilize the 3D spatial features of electrodes. First, the frequency transformation was applied to the EEG signal at each electrode using the fast Fourier transform and the energy value was obtained. Then, the 3D coordinates of the scalp electrodes were projected into the 3D space and the energy values were interpolated using the 3D interpolation method to generate a 3D feature image containing the 3D real spatial location information of the electrodes. Finally, a 3DCNN was designed to match the 3D feature image for feature extraction and recognition. The proposed method obtained 77.19% accuracy in the BCI Competition IV 2a dataset, which is higher than the existing decoding methods. Results from experiments validate the effectiveness of the proposed MI-EEG classification method.

Keywords. Motor imagery; EEG; Fast Fourier Transformation; 3D Interpolation; 3DCNN.

I. Introduction

The brain-computer interface (BCI) enables direct communication between the brain and external devices, completely independent of muscle and peripheral nerve involvement [1,2]. Non-invasive BCI systems have become a hot research topic because of their simplicity, low cost, and non-invasive to the subject [3,4]. Motor imagery electroencephalography (MI-EEG) signals collecting from electrode caps are usually used as the input signal of noninvasive BCI systems, and the subject's motor intention is directly converted into the control signal of external devices by decoding the MI-EEG signals, which are widely used in the fields of disability assistance, rehabilitation medicine, entertainment, and gaming [5-7].

In recent years, many feature extraction algorithms and machine learning classifiers have been applied to decode MI-EEG with good results, such as common spatial pattern (CSP) [8] and principal component analysis (PCA) [9], linear discriminant analysis (LDA), Bayesian classifiers (BCs) and support vector machines (SVM) [10-12]. Although a framework based on traditional feature extraction algorithms and classifiers has been developed, the inherently low signal-to-noise ratio property of MI-EEG [13] and the widespread adoption of manual features directly affect the classification performance of MI-EEG. Thus, how to adaptively extract effective MI-EEG features is a pressing problem to address.

The use of deep learning, particularly convolutional neural networks (CNN) [14], has led to many CNN-based MI-EEG decoding methods being proposed by researchers. In the early CNN-based approaches, researchers used a one-dimensional (1D) kernel to process the raw EEG data directly [15-17] without manually extracting features. However, EEG signals have distinct time-frequency characteristics, and this class of methods is inadequate for extracting the time-frequency features of the signals. At the same time, some other studies hope to identify MI-EEG with the help of CNN for image classification. Tomas et al. used fast Fourier transform (FFT) to calculate the frequency-domain energy of each channel, and the frequency-domain energy spectrum of each electrode was used as a row in the final FFT energy map (FFTEM), and 2D-CNN was used for FFTEM identification [18]. Meanwhile, another time-frequency map recognition method incorporating CNN has been proposed, which uses the short-time Fourier transform (STFT) to obtain the time-frequency image and then performs feature extraction on it with the help of a CNN with a 1D convolution kernel [19-22]. Among the methods for time-frequency feature extraction, the continuous wavelet transform (CWT) is also used in imaging methods. The data acquired by the three leads C3, C4, and Cz were converted into time-frequency maps in [23] by CWT and stacked for imaging input to CNN for identification. These frequency or time-frequency imaging methods combined with CNN gradually improved the classification results. However, the depth information of motor imagery activation has not been reflected, and the precise three-dimensional spatial information of the electrode cannot be encoded into the MI-EEG image, resulting in incomplete spatial information of the electrode and affecting the precision of MI-EEG decoding.

In order to preserve the frequency features of MI-EEG signals, a new EEG classification method was proposed by using the precise spatial features of electrodes to match the CNN spatial convolution capability. Based on FFT and nearest neighbor interpolation (NNI) algorithm, we combine the frequency features with the precise position information of electrodes to generate a series of 3D images. And a matching three-dimensional convolutional neural network (3DCNN) is designed to extract features from the three-dimensional interpolated imaging maps and perform classification.

The rest of this paper is organized as follows. Section 2 provides a detailed description of the 3D feature image interpolation method and structure of the 3DCNN network. The datasets and experimental results used are detailed in section 3. Section 4 is the conclusion of this paper.

II. Methodology

A new imaging method based on the combination of FFT and 3D interpolation algorithms is proposed to make full use of the frequency features of MI-EEG and the 3D position information of electrodes. In this paper, we use the 8-30 Hz frequency band as the frequency band of interest (FOI) for imagining motion [24], which covers the μ and β rhythms, the two most relevant rhythms for the motor imagery task. The method uses FFT to perform feature extraction and derive energy values for frequency features of the FOI. Then, a 3D grid with a fixed dimension size of 32×32×32 is established, and the 3D interpolation is performed based on the NNI algorithm by combining the lead coordinate positions of the BCI acquisition device to obtain a 3D interpolated imaging map, and finally, a 3DCNN is designed for feature extraction and classification of these images.

2.1 Frequency feature calculation

In this paper, the 8-30 Hz frequency band, which is most relevant to the MI task, is used as the imagery motion FOI and uses FFT to extract frequency features. $x_m \in R^{1\times N_s}$ is the MI-EEG signal of the m th electrode in one acquisition experiment, where $m \in \{1,2,3,\ldots, N_c\}$, N_c is the number of electrodes, N_s represents the number of sampling points contained in one acquisition experiment.The MI-EEG data for mth electrode X_m can be described as

$$X_m = [x_m(1), x_m(2), \ldots, x_m(N_s)] \in R^{1\times N_s} \tag{1}$$

Then, divide x_m into N_D windows, $N_D \in N^+$. The data for each window can be expressed as

$$x_{m,j}^D \in R^{1\times\frac{N_s}{N_D}} \tag{2}$$

Where j is the window serial number, $j \in \{1,2,3,\ldots, N_D\}$. So the number of samples contained in $x_{m,j}^D$ is

$$x_{m,j}^D = \left[x_m^{\left((j-1)\times\frac{N_s}{N_D}\right)} \ldots x_m^{\left(j\times\frac{N_s}{N_D}\right)}\right] \in R^{1\times\frac{N_s}{N_D}} \tag{3}$$

FFT is used to convert the time domain signal of each window to the frequency domain. In order to improve the computational frequency resolution, the signal sequence from each window is set to zero at length N_{FFT} , the transformed sequence is scored as $X_{m,j}^F$.

$$X_{m,j}^F \in C^{1\times N_{FFT}} \tag{4}$$

In addition, the 8-30 Hz band sequences corresponding to the FOI is expressed as Equation 5.

$$X_{m,j}^F \in C^{1\times N_F} \tag{5}$$

The length of the frequency band sequence is N_F , the calculation is shown in Equation 6.

$$N_F = (F_H - F_L) \times \frac{N_{FFT}}{\frac{f_s}{2}} \tag{6}$$

where F_H and F_L denote the upper and lower frequency limits of frequency band, and f_s is the sampling frequency.

For each window of the frequency band sequence, the average power is calculated as shown in Eq. (7).

$$X_{m,j}^P = \frac{1}{N_F}\sum X_{m,j}^{F^2} \tag{7}$$

The feature x_m^F is obtained by re-averaging the average power values in the N_D windows, as shown in Eq. (8).

$$x_m^F = \frac{1}{N_D}\sum_{j=1}^{N_D} x_{m,j}^P \tag{8}$$

2.2 3D spatial position interpolation of electrodes

Firstly, a 3D grid is created based on the 3D spatial position coordinates of the electrodes, which are obtained from the electrode distribution map of the BCI system, as shown in Figure 1. Then a 3D grid of size 32×32×32 is created based on the 3D spatial position coordinates of the electrodes. The energy values are interpolated according to the 3D spatial position of the electrodes and the established grid. The frequency features are extended to the 3D space to generate the 3D interpolated imaging map, and the 3D interpolated slice map is shown in Fig. 2.

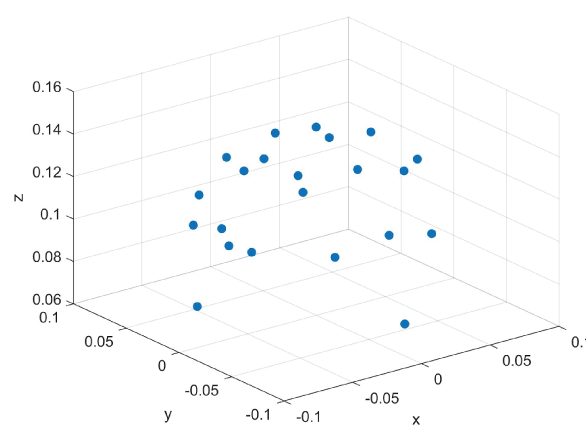

Figure 1. Electrode position in three-dimensional space

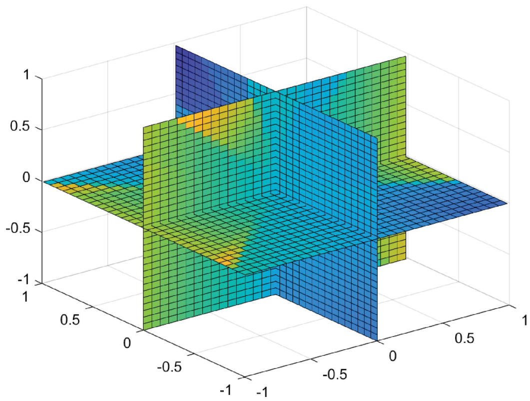

Figure 2. The 3D interpolated slice map

2.3 Structure of 3DCNN

Based on the characteristics of 3D interpolated imaging maps, a 4M3DCNN with a four-module cascade structure is designed for feature matrix extraction and classification. The four-module cascade structure allows the network to extract features from both temporal and spatial dimensions, thereby capturing information encoded in motor imagery across multiple adjacent sampling instants, and the structure diagram of the proposed 4M3DCNN is shown in Figure 3.

Module 1 and Module 2 have the same structure: they contain two 3D convolutional layers and one max-pooling layer with the same size convolutional kernel ($3\times3\times3$) and step size ($3\times3\times3$), and the activation function of both convolutional layers is RELU; Module 3 contains one 3D convolutional layer and one max-pooling layer with a convolutional kernel size of $3\times3\times3$ and a step size of $3\times3\times3$, and the activation function of the convolutional layer is RELU. The activation function of the convolutional layer is RELU, and Module 4 contains two fully connected layers to expand the spatial features extraction and output categories.

All four modules are based on batch normalization (BN and Dropout) technology, with a learning rate of 0.001, a batch size of $32\times32\times1$, a filter size of 32 for the first two convolutional kernels, 64 for the third and fourth filters, and the same Relu activation functions structure for the fifth filter. After five convolutional layers and three maximum pooling layers, two Dense layers are used to avoid the loss of feature information. The final output uses the SoftMax function to obtain the corresponding probability, and the number of outputs depends on the number of classes of motor imagery tasks.

III. Experiments and results

3.1 Dataset description

The dataset used in this study was Dataset 2a from the "BCI Competition 2008" [25], which recorded the EEG signals of nine subjects, collected by 22 electrode caps of the international 10-20 standard and sampled at 250Hz. The data were divided into training and testing sets, with 576 experiments per subject(144 experiments for each type of task, for a total of four types of motor imagery tasks).

3.2 Experiments and results

3.2.1 Experimental results of different network structures

The network structure of 3DCNN is an important factor that affects the decoding accuracy of MI-EEG. In this section, the proposed 3DCNN structure is modified to obtain 2 networks, named M1 and M2, whose detailed structural parameters are shown in Table 1.

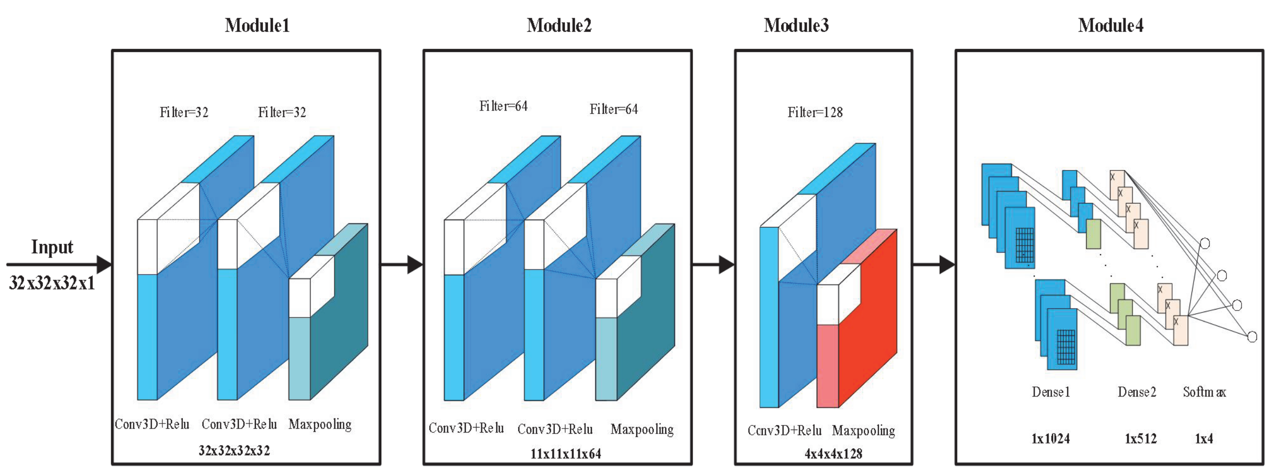

Figure 3.Structure of the designed 4M3DCNN

Table 1 The detailed structures of the 4M3DCNN and its modifications

Model	4M3DCNN	M1	M2
Input:	32×32×32×1	32×32×32×1	32×32×32×1
Module 1	Conv ReLU Max-pool	Conv ReLU Max-pool	Conv ReLU Max-pool
Module 2	Conv ReLU Max-pool	Conv ReLU Max-pool	Conv ReLU Max-pool
Module 3	Conv ReLU Max-pool	Dense ReLU Softmax	Conv ReLU Max-pool
Module 4	Dense ReLU Softmax	-	Conv ReLU Max-pool
Module 5	-	-	Dense ReLU Softmax

M1 and M2 are composed of three and five modules respectively, to justify the number of modules.

The recognition accuracy of different network structures for MI-EEG is shown in Table 2, and it can be observed that the proposed 3DCNN model achieves the best performance for decoding MI-EEG. In contrast, when the 3DCNN model is slightly adjusted, the decoding performance of the modified network for the motor imagery task becomes worse.

Table 2 Results for individual models for BCI Competition IV 2a dataset

Model	Number of modules	Accuracy
4M3DCNN	4	77.19%
M1	3	74.73%
M2	5	66.54%

3.2.2 Comparison results with other methods

The evaluation criterion for the BCI competition IV 2a dataset is ten-fold cross-validation, and the results obtained by the related methods are shown in Table 3.

Under the evaluation criterion of ten-fold cross-validation, the literature [26] used the modified FBCSP to generate novel data as the input of 1DCNN with an accuracy of 74.46%. In the literature [15], the authors used a CNN with two-dimensional convolution to classify FFTEM images and obtained an accuracy of 68% based on 10-fold cross-validation, which verified the feasibility of combining CNN with frequency-space imaging methods; however, its performance still needs to be improved compared with similar methods. In the literature [27], 3D features generated by stacking 2D matrix sequences were used and a multi-branch 3DCNN was designed to extract and identify this feature matrix, which got an accuracy of 75.02%. The improvement of the decoding effect of our proposed method is more significant and achieved higher accuracy than existing methods with 77.19% under the evaluation criteria of tenfold cross-validation.

Table 3 Accuracy of different methods on BCI Competition IV 2a dataset

	Accuracy%			
	FBCSP+ 1DCNN[26]	FFTEM+ 2DCNN[15]	3Ddata+ 3DCNN[27]	Ours
Subject 1	87.50	-	77.40	74.28
Subject 2	65.28	-	60.14	72.25
Subject 3	90.28	-	82.93	78.03
Subject 4	66.67	-	72.29	76.88
Subject 5	62.50	-	75.84	74.57
Subject 6	45.49	-	68.99	79.19
Subject 7	89.58	-	76.04	83.53
Subject 8	83.33	-	76.86	78.03
Subject 9	79.51	-	84.78	77.91
Average	74.46	68.00	75.02	77.19

IV. CONCLUSIONS

The proposed method preserves the depth information of motor imagery activation. The FFT energy value of each electrode is mapped to its equivalent three-dimensional coordinate and three-dimensional interpolation is carried out to avoid the loss of the spatial position information of the electrodes. To fully explore the features of the 3D interpolated imaging map, a 3DCNN is designed, focusing mainly on the unification of the network structure and the physical properties of the MI-EEG. Extensive experimental results on BCI Competition IV 2a show that the 3D interpolated imaging map combined with 3DCNN achieves good results in MI-EEG classification, and it has significant advantages in accuracy compared with existing related methods. Our future research will focus on developing more imaging methods based on 3D interpolation and designing networks with better performance, making it fully applicable to rehabilitation medical and daily life.

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